SMART MANUFACTURING AI

Visual Quality Inspection & Automated Defect Intelligence Platform

RESULT: FAIL

FAILED QUALITY AUDIT

Inspection ID:	INSP-1793378267-195BB7	**Timestamp:**	2026-10-30 22:07:47
Component Name:	Hydraulic component	**Similarity Score (SSIM):**	98.75%
Total Defect Regions:	3	**Affected Surface Area:**	1.503%
High-Severity Defects:	1	**Processing Duration:**	341.9 ms

Visual Evidence & Comparative Computer Vision Analysis

Figure A: Reference (Good Component)

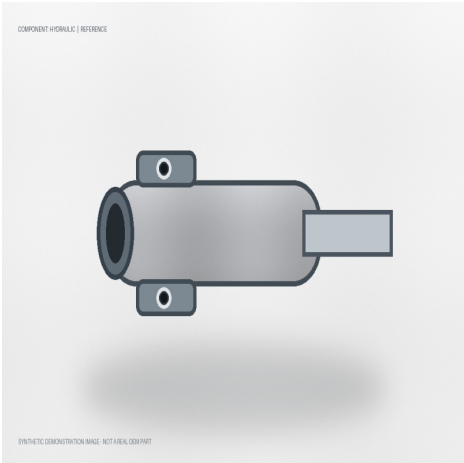


Figure B: Inspected Component

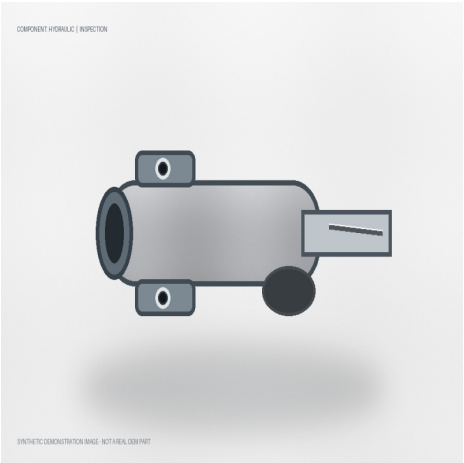


Figure C: SSIM Difference Heatmap

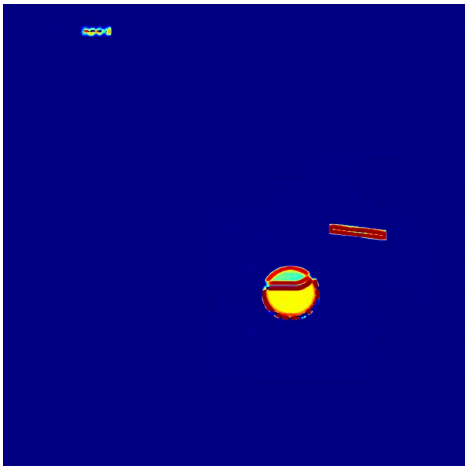
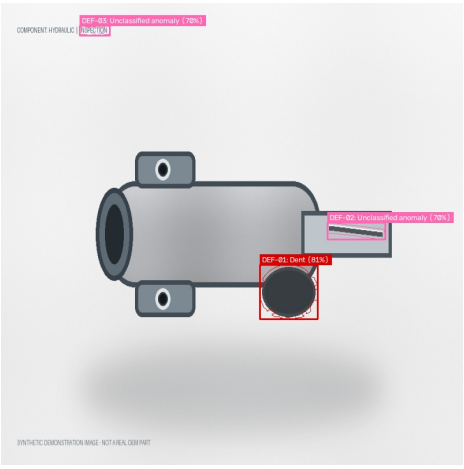


Figure D: Defect Highlighted & Annotated



Detected Defects & Anomaly Registry

Defect ID	Category	Severity	Score	Location (X, Y, W, H)	Area (% / px)	Description

DEF-01	**Dent**	**CRITICAL**	0.81	(448, 453) 100x94	1.12% (7188px)	Localized plastic deformation / depression (1.12% surface area).
DEF-02	**Unclassified anomaly**	**MEDIUM**	0.70	(565, 380) 100x29	0.27% (1751px)	Unclassified photometric anomaly requiring manual inspector verification.
DEF-03	**Unclassified anomaly**	**MEDIUM**	0.70	(136, 39) 52x17	0.11% (682px)	Unclassified photometric anomaly requiring manual inspector verification.

AUTOMATED QUALITY ENGINEERING RECOMMENDATIONS:

- Verify whether surface dent impacts critical mating face sealing or bolt seat flatness.
- Check material handling conveyors and robotic gripper pad pressure to prevent impact indentation.
- Manual inspection recommended: Examine highlighted anomaly area under uniform 1000-lux inspection lighting.

Generated by Smart Manufacturing AI – Computer Vision Quality Inspection Intelligence Suite. Certified for plant floor auditor sign-off.